Joseph K. Miller

Department of Mathematics, Massachusetts Institute of Technology
Mailing address: 100 Memorial Dr, Cambridge, MA 02142, USA
Email: jkm314@mit.edu, jkm314@stanford.edu
Website: hydrodynamical.github.io

Positions

Postdoctoral Associate	Massachusetts Institute of Technology	September 2026 – 2027
SLMath Postdoc	SLMath	Fall, 2025
Szegő Assistant Professor	Stanford University	2024 – 2026

Education

Ph.D. in Mathematics University of Texas, Austin 2018 – September 2024
Advisor: Nataša Pavlović
Thesis: On deriving kinetic equations and their structure from classical many-body dynamics

B.S. in Mathematics University of California, Los Angeles 2015 – 2018
With Honors

Publications

[1] I. Ampatzoglou, J. K. Miller, N. Pavlović, M. Tasković. Inhomogeneous wave kinetic equation and its hierarchy in polynomially weighted L^∞ spaces. *Communications in Partial Differential Equations*, 50(6), 723–765, 2025.

[2] I. Ampatzoglou, J. K. Miller, N. Pavlović, M. Tasković. On the global in time existence and uniqueness of solutions to the Boltzmann hierarchy. *Journal of Functional Analysis*, 289(9), 111079, 2025.

[3] E. Cárdenas, J. K. Miller, N. Pavlović. On the effective dynamics of Bose-Fermi mixtures. *Ars Inveniendi Analytica*, Paper No. 5, 54 pp., 2025. arXiv:2309.04638.

[4] J. K. Miller, A. R. Nahmod, N. Pavlović, M. Rosenzweig, G. Staffilani. A rigorous derivation of the Hamiltonian structure for the Vlasov equation. *Forum of Mathematics, Sigma*, 11, e77, 2023.

[5] I. Ampatzoglou, J. K. Miller, N. Pavlović. A Rigorous Derivation of a Boltzmann System for a Mixture of Hard-Sphere Gases. *SIAM Journal on Mathematical Analysis*, 54(2), 2320–2372, 2022.

[6] J. W. Lyons, J. K. Miller. The Derivative of a Solution to a Second Order Parameter Dependent Boundary Value Problem with a Nonlocal Integral Boundary Condition. *Journal of Mathematics and Statistical Science*, 2015(2), 43–50, 2015.[1]

Preprints

J. K. Miller. Can Language Models Learn a PDE? A Benchmark for Scientific Reasoning through Experimental Design. *Preprint*, 2026. PDF. Code.

J. K. Miller, B. Su. Optimal Graphon Control of Collective Dynamics. *arXiv:2610.31975*, 2026.

J. K. Miller. On Convergence to Equilibrium for the Boltzmann Hierarchy. *arXiv:2610.24089*, 2026. *In submission.*

J. K. Miller. A Formalization of the Mean-Field Derivation of the Vlasov Equation. *arXiv:2607.08986*, 2026.

E. Cárdenas, J. K. Miller, D. Mitrouskas, N. Pavlović. Emergence of fermion-mediated interactions in Bose-Fermi mixtures. *arXiv:2502.18678*, 2025.

[1] This paper was completed while in high school.

In Preparation

I. Ampatzoglou, J. K. Miller, M. Tasković. Scattering for the inhomogeneous Boltzmann equation in weighted L^∞ spaces. *In Preparation.*

Awards and Grants

Frank Gerth III Teaching Excellence Award, University of Texas - Austin	May, 2023.
Leibniz Graduate Student Grant, Oberwolfach Research Institute for Mathematics	May, 2022.
Early Career AMS-NSF-Simons-ICM Travel Grant, AMS	January, 2022.
ICERM Visiting Scholar, ICERM and Brown University	Fall, 2021.
RTG Graduate Fellowship, NSF (DMS-1840314) & University of Texas - Austin	2021-2022.
Provost's Graduate Excellence Fellowship, University of Texas - Austin	2018-2023.
REU, UCLA Department of Mathematics	Summer, 2017.

Upcoming Invited Presentations

Frontier of PDE Formalization and Analysis with AI. Invited speaker, SLMath, Berkeley.	December 7th–8th, 2026.
Correlations and relaxation to equilibrium in kinetic theory. MIT, PDE/Analysis Seminar.	October 27th, 2026.

Conference and Seminar Presentations

Banff International Research Station.	August 16th-21st, 2026.
University of Washington, ICM 2026 Satellite Conference: Recent Developments in the Dynamics of Nonlinear PDEs.	August 3rd-7th, 2026.
MIT, New Challenges in the Derivation and Dynamics of Many Body Systems.	May 9th-10th, 2026.
Stanford University, Plasma Science and Technology Seminar Series.	February 19th, 2026.
Princeton University, Analysis Seminar.	February 16th, 2026.
UC Berkeley, Harmonic Analysis and Differential Equations Student Seminar.	November 25th, 2025.
UC Berkeley, Applied PDE Student Seminar.	October 31st, 2025.
UC Berkeley - Economics Dept., Theory Lunch Seminar.	October 29th, 2025.
SLMath Seminar. (Video Link)	September 19th, 2025.
AMS Special Session: Recent Developments in Nonlinear Evolution Equations.	April 5th-6th, 2025.
Stanford University, Analysis and PDE seminar.	January 28th, 2025.
AMS Special Session: Recent Developments in PDEs and Related Areas.	January 11th, 2025.
AMS Special Session: Advances in Nonlinear Dispersive and Kinetic Equations.	January 10th, 2025.
Stanford University, Faculty Area Research Seminar.	November 21st, 2024.
USC, CAMS Colloquium. (Video Link)	October 28th, 2024.
Penn State, PDE Seminar.	September 18th, 2024.
UC Berkeley, Nonlinear PDEs Summer School.	June 12th-14th, 2024.

UMass - Amherst, Analysis Seminar. April 17th, 2024.

Texas A&M, Texas Analysis and Mathematical Physics Symposium. February 9th-11th, 2024.

Monash University, Conference on Nonlinear dispersive and wave equations. December 10th-15th, 2023.

City University of New York, Harmonic Analysis and PDE seminar. December 1st, 2023.

Kansas State University, 19th Prairie Analysis Seminar. November 3rd-4th, 2023.

Imperial College London, Stability and dynamics in fluid mechanics and kinetic theory. July 10th-14th, 2023.

MIT, NSF Focused Research Group (#2052789). November 19th, 2022.

Emory University, Analysis Seminar. September 15th, 2022.

Summer School on Methods and Models of Kinetic Theory in Pesaro, Italy. June 12th-18th, 2022.

Oberwolfach, Deterministic Dynamics and Randomness in PDE. May 22nd-28th, 2022.

ICERM, Semester on Hamiltonian Methods in Dispersive and Wave Evolution Equations. Fall, 2021.

Institut d'Etudes Scientifiques de Cargèse, School on Mathematical Fluid Dynamics. August 13th-21st, 2021.

University of Texas - Austin, Texas Analysis and Mathematical Physics Symposium. April 9th-11th, 2021.

University of Kentucky, Ohio River Analysis Meeting. March 20th-21st, 2021.

University of California, Davis, Student-Run Analysis & PDE Seminar. February 25th, 2021.

University of Bremen, Nonlinear Dynamics in Quantum Mechanics. October 1st-2nd, 2020.

Upcoming Conferences and Programs

SLMath, AxIOM Program: Singularities & PDEs Representing Natural Phenomena. May 3rd-28th, 2027.

SLMath, AxIOM Program: Machine Learning for Mathematics. March 1st-26th, 2027.

Oberwolfach, Wave-Kinetic Interfaces: Dispersive Dynamics Meets Kinetic Theory. January 17th-22nd, 2027.

Canadian Mathematical Society Winter Meeting, Montréal. December 11th-14th, 2026.

Simons Collaboration on Wave Turbulence Annual Meeting. December 3rd-4th, 2026.

Other Conferences and Programs Attended

MIT Mathematics Department Retreat, New Hampshire. October 16th-18th, 2026.

UMass - Amherst, NSF Focused Research Group Annual Meeting. June 25th-26th, 2025.

ICERM, Interacting Particle Systems: Analysis, Control, Learning and Computation. May 6th-10th, 2024.

UNC - Chapel Hill, Mini-school on Recent Developments in Wave Turbulence Theory. April 5th-6th, 2024.

ETH Zürich, Kinetic and Hydrodynamic PDEs. January 16th-19th, 2024.

Simons Collaboration on Wave Turbulence Annual Meeting. November 30th-December 1st, 2023.

MIT, Summer School on Wave Turbulence. July 24th-28th, 2023.

Simons Collaboration on Wave Turbulence Annual Meeting. December 1st-2nd, 2022.

Banff International Research Station, Recent Progress in Kinetic and Integro-Differential Equations. November 6th-11th, 2022.

ICERM, Workshop on Probabilistic Methods in Geometry and Analysis. October 17th-21st, 2022.

Simons Collaboration on Wave Turbulence Annual Meeting. December 2nd-3rd, 2021.

MSRI, CRM-PIMS Summer School in Probability (virtual). May 24th-June 18th, 2021.

University of Texas - Austin, Summer Program in PDEs (Teaching Assistant, virtual). May 17th-21st, 2021.

École Normale Supérieure de Lyon, Scaling Limits in Kinetic Theory. September 30th-October 4th, 2019.

University of Texas - Austin, NSF-CBMS Conference: Fitting Smooth Functions to Data. August 5th-9th, 2019.

Universitat de Barcelona, Barcelona Analysis Conference. June 25th-28th, 2019.

Kácov, EMSSAM: Mathematical Aspects of Fluid Flows. May 26th-31st, 2019.

Teaching Experience

Instructor	Math 171, Stanford University	Spring, 2026.
Instructor	Math 20, Stanford University	Winter, 2026.
Instructor	Math 51, Stanford University	Winter, 2025.
Instructor	Math 171, Stanford University	Fall, 2024.
Instructor	Real Analysis Prelim Bootcamp, UT - Austin	Summer, 2024.
Teaching Assistant	M361K, UT - Austin	Spring, 2022.
Workshop Assistant	RTG Workshop on Analysis & PDEs, UT - Austin	Summer, 2021.
Instructor	Real Analysis Prelim Bootcamp, UT - Austin	Summer, 2020.
Undergraduate Mentor	Directed Reading Projects, UT - Austin	Fall, 2020 - Spring, 2021.
Teaching Assistant	Los Angeles Math Circle, UCLA	Fall, 2017 - Spring, 2018.

Professional Service

Referee for *Memoirs of the American Mathematical Society.*

Referee for *Journal of Statistical Physics.*

Referee for *SIAM Journal on Mathematical Analysis.*